



HK IMO Training 2023-2024
Problem Set 11
For 23 December 2023



Problem 41. Let n be a positive integer. Find the largest possible value of

$$S = x_1x_2x_3 + x_4x_5x_6 + \cdots + x_{3n-2}x_{3n-1}x_{3n} + x_{3n+1}$$

where $(x_1, x_2, \dots, x_{3n+1})$ is a permutation of $(1, 2, \dots, 3n+1)$.

Problem 42. Let $n \geq 3$. Two players X and Y play a game in a regular n -gon with a vertex A which is assigned as a trap. Initially, X places a token at one of the vertices other than A . In each round, Y announces a positive integer k (not necessarily the same each time). Then X moves the token by k vertices in the clockwise or anticlockwise direction. Find all n for which X has a strategy that guarantees the token never falls into the trap regardless of how Y plays.

Problem 43. In acute $\triangle ABC$, let D be the foot of altitude from A . Let E be a point on the altitude from C such that $BE \perp BC$, and let F be a point on the altitude from B such that $CF \perp BC$. Prove that $AD = BC$ if and only if A, E, D, F are concyclic.

Problem 44. Let p be a prime number such that $p \equiv 1 \pmod{7}$. Prove that there exists a positive integer n such that $p \mid n^3 + n^2 - 2n - 1$.